

Differentiation – Product Rule: Worked Examples

Product Rule

If $y = uv$ where u and v are functions of x , then

$$\frac{dy}{dx} = u \frac{dv}{dx} + v \frac{du}{dx}$$

Example

Differentiate:

$$y = (4x^{\frac{3}{2}})(\tan x)$$

$$u = 4x^{\frac{3}{2}}; \quad \frac{du}{dx} = \left(4 \left(\frac{3}{2}\right) x^{\frac{1}{2}}\right) = 6x^{\frac{1}{2}}$$

$$v = \tan x; \quad \frac{dv}{dx} = \sec^2 x$$

$$\frac{dy}{dx} = \left(4x^{\frac{3}{2}}\right)(\sec^2 x) + (\tan x)(6x^{\frac{1}{2}})$$

$$\frac{dy}{dx} = 4x^{\frac{3}{2}}\sec^2 x + 6x^{\frac{1}{2}}\tan x$$

$$\frac{dy}{dx} = 4\sqrt{x^3}\sec^2 x + 6\sqrt{x}\tan x$$

This could be factorised to

$$\frac{dy}{dx} = 2\sqrt{x}(2x\sec^2 x + 3\tan x)$$